

SUNNY KUSHWAHA

Senior Lead Engineer | Architecting Production-Scale AI-Powered Distributed Systems | GenAI (RAG, LLMs) | Azure & AWS | Microservices

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🔗 <https://github.com/sunny-dev007> 📍 Pune, INDIA

☆ GenAI & Architecture Portfolio: <https://workwithcopilot.azurewebsites.net>

SUMMARY

Senior Lead Engineer with strong expertise in cloud architecture, microservices, and GenAI-powered platforms. Proven track record in designing scalable distributed systems on AWS and Azure and building enterprise AI solutions using Azure OpenAI.

Led end-to-end development of AI-driven platforms, including document intelligence and conversational systems, improving operational efficiency and reducing manual effort.

Recognized for improving deployment efficiency by 50% through CI/CD automation and winning the Prompt Master award for delivering innovative GenAI workflows.

Focused on building scalable, high-performance cloud-native systems that drive enterprise innovation and intelligent automation.

EXPERIENCE

Senior Lead Engineer

04/2023

Nitor infotech Pvt Ltd.

- Architected high-concurrency apps using microservices, event-driven patterns.
- Architected and implemented scalable microservices-based systems using Node.js on AWS and Azure, supporting high-concurrency enterprise applications.
- Developed and developed GenAI platforms leveraging Azure OpenAI and vector databases for enterprise-grade document intelligence and conversational AI systems.
- Built AI-driven enterprise solutions including document intelligence systems and contextual QnA assistants integrated with knowledge platforms.
- Defined system architecture ensuring high availability and low latency through API gateway design, caching strategies, and asynchronous processing.
- Implemented end-to-end cloud architecture on Azure, utilizing services like App Services, Azure SQL, and secure networking.
- Designed CI/CD pipelines with GitHub Actions and Terraform for automated, scalable, and cost-efficient deployments.
- Recognized by client leadership for successful delivery of critical releases and business solutions leveraging AI innovation.

Senior Software Engineer

12/2019 - 04/2023

Infopro Learning

- Led end-to-end feature development including requirement analysis, system design, database architecture, and deployment.
- Incorporated third-party APIs and services, ensuring reliable data exchange and system interoperability.
- Developed back-end components, connected the application with other web services.
- Built a serverless application using AWS Services like DynamoDB, Lambda functions with Node.js, and API Gateway.
- Architected and developed scalable web applications using Node.js, React.js, and Python, supporting high-concurrency user workloads across distributed systems.
- Designed and implemented microservices architecture on Azure and AWS, utilizing services like Azure Functions, AWS Lambda, and API Gateway, to improve system scalability and performance under high-load conditions.
- Developed reusable backend components and RESTful APIs, enabling seamless integration across platforms and improving data flow efficiency and system performance.
- Led system design and implementation, including database architecture and CI/CD via Azure DevOps. Enhanced efficiency and enabled scalable deployments.



SKILLS

Backend & Frontend Development:

Node.js · Backend Development · Fast API · Typescript · JavaScript · Express · API Design · React.js · GraphQL · Python · PostgreSQL

Architecture & Design:

Solid Principle · System Design · Design Patterns · Microservices Architecture · Distributed Systems · High Availability · Scalability · Event-Driven Architecture

Data & Messaging:

Event Streaming Platforms · Redis · Kafka · RabbitMQ · Data Partitioning & Sharding

Cloud & DevOps:

AWS Lambda · API Gateway · DynamoDB · S3 · Azure App Services · Azure Functions · Blob Storage · Azure SQL · Docker · Kubernetes · CI/CD · GitHub Actions · SQL

GenAI & Machine Learning:

Azure OpenAI · Generative AI · Azure AI Studio · Azure AI Foundry · OpenAI APIs · LLMs · RAG · AI Agent Architecture · LLM Orchestration (LangChain, LangGraph) · Vector Databases (Weaviate, Pinecone, FAISS) · Embeddings & Semantic Search · AI Workflow Automation · LLM Evaluation & Guardrails · Cost & Latency Optimization for LLMs

EXPERIENCE

Software Engineer 08/2015 - 12/2019
Marketgoal Digital Solutions Pvt Ltd

- Developed and maintained scalable and secure web applications using PHP frameworks (Laravel, CodeIgniter), MySQL, and integrated third-party APIs.
- Built reusable backend components and RESTful APIs, ensuring seamless integration with frontend and external systems.
- Led end-to-end feature development including requirement analysis, database design, development, and deployment.
- Created a serverless application using AWS Services such as DynamoDB, Lambda, and API Gateway with Node.js.

PROJECTS

Document Intelligence & Research Platform 06/2023 - Present

- Role: Project Lead | Tech: Azure, Node.js, Python, Azure OpenAI, Weaviate
- Architected and delivered a scalable GenAI-powered Document Intelligence platform on Azure, enabling enterprise research workflows using Retrieval-Augmented Generation (RAG)
 - Designed end-to-end architecture integrating Azure SQL, Blob Storage, Azure OpenAI (LLMs & embeddings), and Weaviate for hybrid semantic search (BM25 + vector)
 - Built document processing pipeline with chunking, embedding generation, and vector indexing, enabling efficient multi-document retrieval and contextual summarization
 - Developed GenAI workflows including key-findings summarization, contextual Q&A, and SOW-to-survey generation using fine-tuned LLM models
 - Designed conversational AI agents for survey intake and IDI (interview) simulation, improving research automation and decision-making
 - Implemented scalable AI pipelines with LLM orchestration, prompt engineering, and cost optimization strategies
 - Led architecture decisions, system design, and cross-functional collaboration, delivering high-impact AI-driven solutions

AI-Driven DevOps Migration & Cost Optimization Platform 08/2024 - Present

- Role: Project Lead | Tech: Azure, AWS, Node.js, Python, Azure OpenAI, Terraform, Kubernetes, CI/CD
- Led migration of legacy apps to cloud-native microservices on Azure and AWS.
 - Improved system scalability, reliability, and deployment efficiency across cloud-native environments.
 - Designed AI-driven cost optimization platform using Azure OpenAI to analyze cloud usage patterns, detect anomalies, and recommend cost-saving strategies (FinOps).
 - Built intelligent DevOps pipeline optimization system leveraging LLMs to analyze CI/CD workflows, identify bottlenecks, and automate pipeline improvements.
 - Implemented Infrastructure as Code (Terraform) and automated CI/CD pipelines using GitHub Actions, reducing deployment time and manual intervention by over 50%.
 - Designed Kubernetes-based containerized architecture enabling scalable deployments, auto-scaling, and efficient resource utilization.
 - Developed AI-powered insights dashboard for monitoring cloud costs, usage trends, and optimization recommendations across multi-cloud environments.
 - Applied event-driven architecture and monitoring to ensure high availability, observability, and fault tolerance across distributed systems.
 - Led cross-functional teams and collaborated with stakeholders, driving architecture decisions, cloud migration strategy, and enterprise-scale DevOps transformation.

Scalable Full-Stack Learning & Content Platform 02/2021 - 02/2023

- Role: Senior Software Engineer | Tech: Node.js, React.js, AWS/Azure, MySQL
- Designed and implemented event-driven microservices architecture using Node.js and AWS services for high-throughput and low-latency systems.
 - Built asynchronous communication workflows using Kafka and message queues, enabling scalable and decoupled service interactions.
 - Implemented caching strategies using Redis to reduce latency and improve system performance under high load.
 - Designed API Gateway-based routing, authentication, and rate limiting to ensure secure and scalable API management.
 - Developed fault-tolerant systems with retry mechanisms, circuit breakers, and monitoring for high availability.
 - Optimized system scalability using horizontal scaling, load balancing, and database partitioning strategies.

KEY ACHIEVEMENTS

- 🔹 Accelerated Deployment & Engineering Efficiency
Improved deployment efficiency by 50% through automated CI/CD pipelines and infrastructure optimization.
- 🔹 Recognized as "Prompt Master" for AI Innovation
Won organization-wide PROMPT-MASTER competition by designing and demonstrating practical GenAI-driven workflows. Showcasing real-world applications of GenAI to solve business problems and improve operational efficiency
- 🔹 Delivered GenAI-powered Enterprise Solutions
Designed and implemented AI-driven platforms using Azure OpenAI, RAG architecture, and vector databases, significantly improving operational efficiency and reducing manual workload.
- 🔹 Recognized for Architectural Ownership & Delivery Excellence
Recognized by client and leadership for driving end-to-end ownership, making key architectural decisions, and successfully delivering critical production releases under tight timelines.

QUALIFICATIONS

- MCA (Regular)
IEC college of engineering and technology
07/2010 - 07/2013 Gr. Noida (UP)
- PGDCA (Diploma)
Government Polytechnic, Kanpur
07/2008 - 07/2010 Kanpur (UP)
- B.Sc. (Mathematics)
Bundelkhand University
07/2005 - 06/2008 Jhansi (UP)
- Intermediate (12th)
UP Board
06/2003 - 07/2005 Jhansi (UP)
- High School (10th)
UP Board
2003 - 2003 Jhansi (UP)